

Week-6-PP

September 6, 2026

1 Interactive Bar Chart with Plotly

1.1 Week 6 Practice Program

1.1.1 Problem Statement

Create an interactive bar chart showing the population of different cities.

1.1.2 Dataset

The dataset contains the population of six major Indian cities in lakhs.

```
[2]: import pandas as pd
import plotly.express as px
```

```
[3]: data = {
    "City": [
        "Mumbai",
        "Delhi",
        "Bengaluru",
        "Hyderabad",
        "Chennai",
        "Kolkata"
    ],
    "Population (Lakhs)": [
        204,
        167,
        136,
        108,
        115,
        149
    ]
}
```

```
[5]: df = pd.DataFrame(data)

df
```

```
[5]:
```

	City	Population (Lakhs)
0	Mumbai	204
1	Delhi	167
2	Bengaluru	136
3	Hyderabad	108
4	Chennai	115
5	Kolkata	149

```
[6]: print("Dataset Shape:", df.shape)

print("\nCities:")
print(df["City"].tolist())

print("\nPopulation:")
print(df["Population (Lakhs)"].tolist())
```

Dataset Shape: (6, 2)

Cities:

['Mumbai', 'Delhi', 'Bengaluru', 'Hyderabad', 'Chennai', 'Kolkata']

Population:

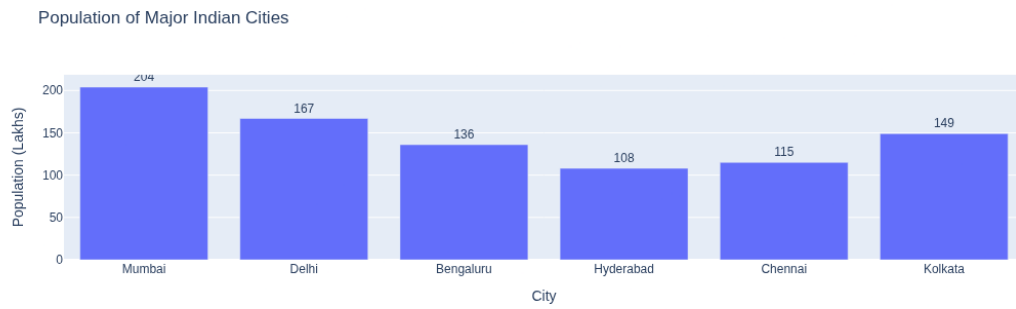
[204, 167, 136, 108, 115, 149]

```
[8]: fig = px.bar(
    df,
    x="City",
    y="Population (Lakhs)",
    title="Population of Major Indian Cities",
    text="Population (Lakhs)"
)

fig.update_traces(
    textposition="outside"
)

fig.update_layout(
    xaxis_title="City",
    yaxis_title="Population (Lakhs)"
)

fig.show()
```



```
[9]: sorted_df = df.sort_values(
      by="Population (Lakhs)",
      ascending=False
    )

sorted_df
```

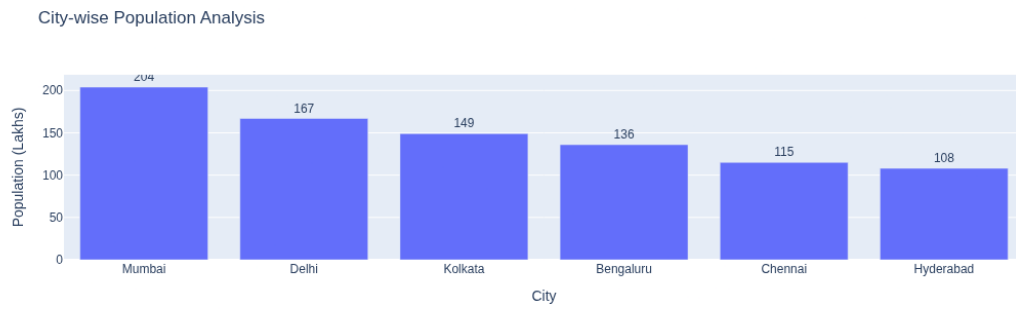
```
[9]:      City  Population (Lakhs)
0    Mumbai                204
1     Delhi                167
5    Kolkata                149
2  Bengaluru                136
4     Chennai                115
3  Hyderabad                108
```

```
[10]: fig = px.bar(
      sorted_df,
      x="City",
      y="Population (Lakhs)",
      title="City-wise Population Analysis",
      text="Population (Lakhs)"
    )

fig.update_traces(
    textposition="outside"
)

fig.update_layout(
    xaxis_title="City",
    yaxis_title="Population (Lakhs)"
)

fig.show()
```



```
[11]: highest = df.loc[
        df["Population (Lakhs)"].idxmax()
    ]

    print("City with Highest Population:")
    print(highest)
```

```
City with Highest Population:
City                Mumbai
Population (Lakhs)      204
Name: 0, dtype: object
```

```
[12]: lowest = df.loc[
        df["Population (Lakhs)"].idxmin()
    ]

    print(
        "Lowest Population:",
        lowest["City"],
        "-",
        lowest["Population (Lakhs)"],
        "Lakhs"
    )
```

```
Lowest Population: Hyderabad - 108 Lakhs
```

```
[13]: average_population = df["Population (Lakhs)"].mean()

    print(
        "Average Population:",
        round(average_population, 2),
        "Lakhs"
    )
```

```
Average Population: 146.5 Lakhs
```

```
[14]: print("CITY POPULATION ANALYSIS")
      print("-" * 35)

      print("Total Cities:", len(df))

      print(
          "Highest Population:",
          df.loc[df["Population (Lakhs)"].idxmax(), "City"]
      )

      print(
          "Lowest Population:",
          df.loc[df["Population (Lakhs)"].idxmin(), "City"]
      )

      print(
          "Average Population:",
          round(df["Population (Lakhs)"].mean(), 2),
          "Lakhs"
      )
```

CITY POPULATION ANALYSIS

Total Cities: 6

Highest Population: Mumbai

Lowest Population: Hyderabad

Average Population: 146.5 Lakhs

```
[15]: fig.write_html(
      "city_population_chart.html"
      )

      print("Interactive chart saved successfully.")
```

Interactive chart saved successfully.

```
[16]: df.to_csv(
      "city_population.csv",
      index=False
      )

      print("Dataset saved successfully.")
```

Dataset saved successfully.

```
[ ]:
```